Dale T. Clement, PhD

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SKILLS

Subject-matter expertise: Quantitative ecology, conservation science, evolution, epidemiology, statistics, applied math

Communication: Scientific writing (8 peer-reviewed publications & 2 preprints), Peer review (8 articles, 1 grant), verbal communication (13 conference presentations), grant writing

Leadership: Project management (8 projects), mentorship (4 mentees), international collaboration (6 publications)

Technical Skills: Population viability analysis, demographic modeling, agent-based simulation, maximum likelihood & Bayesian inference, spatial statistics, machine learning, uncertainty analysis

Coding: R (tidyverse, sf, sp, ggplot2), C++, bash scripting, high-performance computing ∗ **Workflow**: Git, Quarto

EDUCATION

PhD, Population Biology, University of California - Davis Sep 2017 – Aug 2022
Dissertation: Modeling the effects of evolutionary history and stochasticity on population responses to rapid environmental change

BA, Biology and Mathematics, Dartmouth College Sep 2013 – Jun 2017
Magna cum laude with High Honors in Biology

EXPERIENCE

Postdoctoral Researcher Sep 2024 – Current
Nicholas Kortessis Lab, Department of Biology, Wake Forest University *Winston-Salem, NC*

- Developed spatially-explicit infectious disease models integrating human mobility patterns via cell phone and survey data, identifying circumstances in which changes in travel behavior increase disease transmission.
- Partnered with the Forsyth County, NC, Department of Public Health (DPH) to analyze the **spatiotemporal patterns** of mosquito abundance and West Nile Virus (WNV) using **species distribution modeling**, combined with land use, climate, eBird, and mosquito trap data, with the aim of identifying priority trapping locations to maximize WNV detection.
- Delivered monthly **updates to stakeholders** (Forsyth DPH) via in-person meetings and **Quarto-generated reports**.

Postdoctoral Research Associate Sep 2022 – Aug 2024
Andrew Storfer Lab, School of Biological Sciences, Washington State University *Pullman, WA*

- Used **agent-based simulations** to conduct **population viability analysis** for the endangered Tasmanian devil (*Sarcophilus harrisii*), incorporating evolution to yield more accurate extinction risk estimates than purely demographic models.
- Evaluated the impact of **conservation intervention** on Tasmanian devil populations, including rigorous quantification of **model uncertainty**, demonstrating that translocation would have only minor effects on long-term devil abundance.
- **Validated models** with Approximate Bayesian Computation using genomic, demographic, and disease prevalence data, confirming that model predictions were consistent with observed population patterns.
- Contributed statistical analysis and feasibility assessment to **three grant proposals** and reports and **mentored** two graduate students in quantitative methods.

National Science Foundation Graduate Research Fellow Sep 2017 – Aug 2022
Sebastian Schreiber Lab, Center for Population Biology, University of California, Davis *Davis, CA*

- Modeled the effect of evolution and plasticity on natural populations' ability to cope with environmental change, identifying which types of environmental change are most harmful to populations with different evolutionary histories.
- Funded research through an NSF graduate research fellowship **($102,000**), a Dean's Distinguished graduate fellowship **($27,000**), and a UC Davis graduate student assembly travel award ($500).
- Served in student leadership for four years, **mediating between departmental stakeholders**, drafting recommendations, and mentoring fellow students.